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Self-excited oscillations and limit cycles

Lecture content

4.1 Weakly damped free vibration

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4.3 Amplitude selection

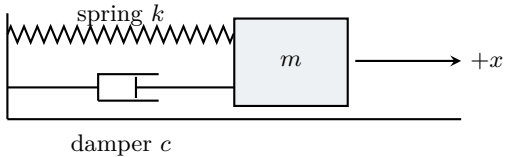
4.4 Stability and limit cycles

4.5 Introduction to Hopf bifurcation

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Release the mass: what will damping change?

Pull the mass away from equilibrium and release it from rest. The spring reverses its motion; the damper makes successive peaks smaller.



x: displacement from equilibrium (m); m: mass (kg); k: stiffness (N/m); $c > 0$: viscous damping coefficient (N s/m). t: time (s); dots denote d/dt.

$$\underbrace{m\ddot{x}}_{\text{mass times acceleration}} = \underbrace{-kx}_{\text{spring force}} \underbrace{-c\dot{x}}_{\text{damper force}}, \qquad x(0) = R, \quad \dot{x}(0) = 0.$$

$R > 0$ is release displacement (m). We want to calculate how fast the peaks shrink and whether the oscillation frequency also changes.

Weak damping separates oscillation and decay

Divide Newton's equation by m and introduce two useful quantities:

$$\omega_0 = \sqrt{k/m}, \qquad \zeta = \frac{c}{2m\omega_0}, \qquad \ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2 x = 0.$$

What do these quantities mean?

ω_0 is the undamped natural angular frequency (rad/s); ζ is the dimensionless damping ratio.

When $\zeta \ll 1$, the loss per cycle is small. The displacement oscillates rapidly while its peak amplitude $a(t)$ (m) changes more slowly.

$$x(t) \approx a(t)\cos\omega_0 t.$$

The cosine describes motion within a cycle. The function $a(t)$ describes how the peaks evolve over many cycles. We will obtain $a(t)$ from energy loss.

Why multiply the equation by velocity?

Force times velocity is mechanical power. Multiplying the equation by $\dot{x}$ therefore connects the forces to energy transfer:

$$m\ddot{x}\dot{x} + kx\dot{x} = -c\dot{x}^2.$$

Each term on the left is a time derivative:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}m\dot{x}^2\right) = m\dot{x}\ddot{x}, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}kx^2\right) = kx\dot{x}.$$

$$\boxed{E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2, \qquad \dot{E} = -c\dot{x}^2 \leq 0.}$$

This is exact. The damper removes energy in either direction of travel. At a turning point $\dot{x} = 0$, instantaneous power loss is zero.

Connect energy loss to shrinking amplitude

During one weakly damped cycle, a changes little. To leading order,

$$x \approx a\cos\omega_0 t, \qquad \dot{x} \approx -a\omega_0 \sin\omega_0 t.$$

Using $m\omega_0^2 = k$, kinetic plus potential energy becomes

$$E \approx \tfrac{1}{2}ka^2(\sin^2\omega_0 t + \cos^2\omega_0 t) = \tfrac{1}{2}ka^2, \qquad \dot{E} \approx ka\dot{a}.$$

Average the loss over a cycle: $\langle\cdot\rangle$ denotes this average, and $\langle\sin^2\omega_0 t\rangle = 1/2$. Hence

$$ka\dot{a} = -\tfrac{1}{2}ca^2\omega_0^2 \quad \Rightarrow \quad \boxed{\dot{a} = -\frac{c}{2m}a = -\zeta\omega_0 a.}$$

For $a > 0$, divide by ka. The amplitude loses the same fraction per unit time, rather than a fixed number of metres per second.

Solve the slow amplitude decay

Separate variables and integrate from $a(0) = R$:

$$\int_R^{a(t)} \frac{\mathrm{d}a}{a} = -\zeta\omega_0 \int_0^t \mathrm{d}t \quad \Longrightarrow \quad \ln\frac{a(t)}{R} = -\zeta\omega_0 t.$$

$$\boxed{a(t) = Re^{-\zeta\omega_0 t}.}$$

The leading waveform is $x \approx a(t)\cos\omega_0 t$. Its derivative is

$$\dot{x} \approx \dot{a}\cos\omega_0 t - a\omega_0 \sin\omega_0 t, \qquad \dot{x}(0) \approx -\zeta\omega_0 R.$$

Release from rest requires $\dot{x}(0) = 0$, but the approximate cosine predicts $-\zeta\omega_0 R$. Relative to the velocity scale $\omega_0 R$, the mismatch has size ζ: halving ζ halves this error.

The exact solution adds a sine term. At $t = 0$ that term has zero displacement but contributes velocity $+\zeta\omega_0 R$, cancelling the mismatch. The next slide finds its coefficient from $\dot{x}(0) = 0$.

The exact solution must satisfy both release conditions

For $\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2 x = 0$, try $x = e^{rt}$, where r is a rate (s^{-1}):

$$r^2 + 2\zeta\omega_0 r + \omega_0^2 = 0 \Rightarrow r = -\zeta\omega_0 \pm i\omega_d, \qquad \omega_d = \omega_0\sqrt{1-\zeta^2}.$$

$i^2 = -1$; ω_d is the damped angular frequency. For $0 < \zeta < 1$, constants C_1, C_2 (m) give

$$x = e^{-\zeta\omega_0 t}(C_1 \cos\omega_d t + C_2 \sin\omega_d t).$$

$$x(0) = R \Rightarrow C_1 = R, \qquad \dot{x}(0) = 0 \Rightarrow -\zeta\omega_0 R + \omega_d C_2 = 0.$$

$$\boxed{x = Re^{-\zeta\omega_0 t}\left[\cos\omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_d t\right].}$$

The sine term leaves the initial displacement unchanged and supplies the velocity needed to make $\dot{x}(0) = 0$.

Damping changes frequency less strongly than amplitude

The exact damped frequency is $\omega_d = \omega_0\sqrt{1-\zeta^2}$. For a small quantity z, the Taylor expansion gives

$$\sqrt{1-z} = 1 - \frac{z}{2} + O(z^2).$$

Set $z = \zeta^2$, then subtract ω_0 and divide by it:

$$\omega_d \approx \omega_0\left(1 - \frac{\zeta^2}{2}\right), \qquad \boxed{\frac{\omega_d - \omega_0}{\omega_0} \approx -\frac{\zeta^2}{2}.}$$

At $\zeta = 0.02$, the relative decrease is $0.02^2/2 = 0.0002$, or 0.02%. Halving ζ makes this decrease four times smaller.

Amplitude decay has rate $\zeta\omega_0$, proportional to ζ. The leading frequency change is proportional to ζ^2. Peaks can shrink noticeably while the period changes very little.

Weak damping: a numerical example

Take $m = 1$ kg, $k = 100$ N/m, $c = 0.40$ N s/m, and $R = 0.10$ m.

$$\omega_0 = \sqrt{100/1} = 10 \text{ rad/s}, \qquad \zeta = \frac{0.40}{2(1)(10)} = 0.020.$$

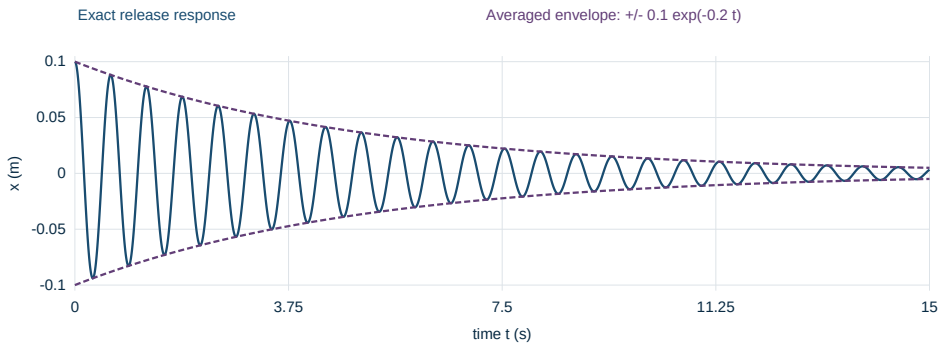
$$a(t) = 0.10e^{-0.20t} \text{ m}, \qquad a(5) = 0.10e^{-1} = 0.03679 \text{ m}.$$

Let $t_{1/2}$ be the half-amplitude time and T the leading period. For half the initial amplitude, $e^{-0.20t} = 1/2$:

$$t_{1/2} = \frac{\ln 2}{0.20} = 3.466 \text{ s}, \qquad T = \frac{2\pi}{\omega_0} = \frac{2\pi}{10} = 0.6283 \text{ s}.$$

Amplitude halves over about $3.466/0.6283 = 5.52$ cycles. The oscillation period and the decay time are different quantities.

Fast oscillation inside a slowly shrinking envelope



The blue response satisfies the exact release conditions. The dashed curves show the averaged amplitude $\pm a(t)$; the small difference is higher order in weak damping.

What does negative damping mean physically?

An ordinary damper opposes velocity. A powered force can instead reinforce it:

	Force	Power delivered to the mass	
Passive damping	$-c\dot{x}$	$-c\dot{x}^2 \le 0$	$(c > 0)$.
Active assistance	$+c\dot{x}$	$+c\dot{x}^2 \ge 0$	

Putting an assisting force on the left of Newton's equation gives

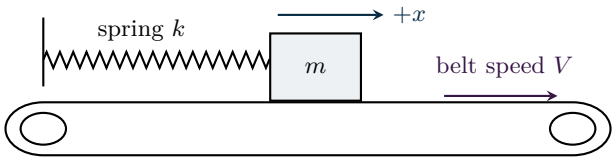
$$m\ddot{x} + kx = c\dot{x} \quad \Rightarrow \quad m\ddot{x} - c\dot{x} + kx = 0.$$

The velocity coefficient is negative, so a disturbance gains energy. A motor or another power source must supply that energy. Negative damping is not an energy source by itself.

The moving-belt example shows how this effect can arise without a controller prescribing a periodic force.

Negative damping: compare rightward and leftward motion

A motor drives the belt right at speed V (m/s). At rest, the spring balances the rightward friction force.



x: displacement from this equilibrium (m); m: mass (kg); k: spring stiffness (N/m). Assume $|\dot{x}| < V$ and friction that decreases as sliding speed increases.

Block moving right: $\dot{x} > 0$

The belt slides past it more slowly. Friction becomes stronger than its equilibrium value: the extra rightward force assists the motion.

Block moving left: $\dot{x} < 0$

The belt slides past it faster. Friction becomes weaker: relative to equilibrium, the force change is leftward and assists the leftward motion.

Why weaker rightward friction can assist leftward motion

Let $F_b(s) > 0$ be the rightward friction force (N), with sliding speed $s = V - \dot{x}$ (m/s). At rest, $s = V$.

Measure force changes relative to the friction-balanced equilibrium:

$$\Delta F = F_b(V - \dot{x}) - F_b(V).$$

An illustrative comparison at the same displacement

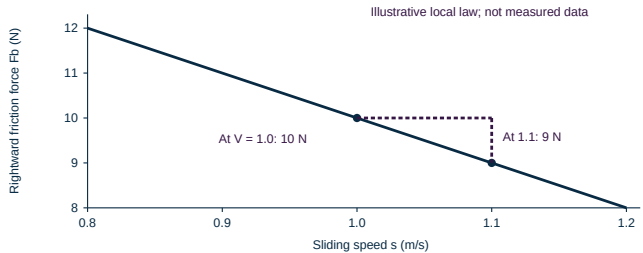
Suppose equilibrium friction is 10 N. If motion right raises it to 11 N, the extra force is +1 N. If motion left lowers it to 9 N, the extra force is −1 N.

$\dot{x}$ (m/s)	ΔF (N)	$\Delta F\dot{x}$ (W)
+0.10	+1	+0.10
−0.10	−1	+0.10

Actual friction remains rightward in both cases. Its change from equilibrium reinforces the disturbance. The belt motor supplies the energy; friction is not creating energy.

Read a decreasing friction curve

The horizontal variable is sliding speed $s = V - \dot{x}$ (m/s). The vertical variable $F_b(s)$ is the rightward friction force (N). Assume the friction law decreases with s near $s = V$.



Read from left to right: sliding speed increases, while friction force decreases. That is what a decreasing curve means. It is an assumption about this operating region, not a property of every friction law.

Why is the derivative of that curve negative?

A derivative is a local slope: vertical change divided by horizontal change. For the illustrative graph at $V = 1.0$ m/s,

$$\Delta s = 1.1 - 1.0 = +0.1 \text{ m/s}, \qquad \Delta F_b = 9 - 10 = -1 \text{ N}.$$

$$\frac{\Delta F_b}{\Delta s} = \frac{-1}{+0.1} = -10 \text{ N s/m}.$$

The example is a straight segment, so this is also its derivative. For a curved law, shrink the interval to obtain the tangent slope:

$$\boxed{F_b'(V) = \left.\frac{\mathrm{d}F_b}{\mathrm{d}s}\right|_{s=V} < 0.}$$

The prime here means differentiation with respect to sliding speed s, evaluated at V. It is not a time derivative. When the block moves right, $s = V - \dot{x}$ decreases, so we move left on this graph and friction increases.

Subtract equilibrium before identifying damping

The coordinate x measures departure from the friction-balanced position. An additional displacement x changes the spring force by $-kx$.

The unchanging equilibrium spring and friction forces cancel. Newton's equation for the departure is therefore

$$m\ddot{x} = -kx + \underbrace{F_b(V - \dot{x}) - F_b(V)}_{\Delta F}.$$

Approximate the force change for small velocity

The argument of F_b changes from V by $-\dot{x}$. A first-order Taylor expansion gives

$$F_b(V - \dot{x}) = F_b(V) + F_b'(V)(-\dot{x}) + O(\dot{x}^2).$$

$$\Delta F \approx -F_b'(V)\dot{x}, \qquad \boxed{m\ddot{x} + F_b'(V)\dot{x} + kx = 0.}$$

$F_b'(V) = \mathrm{d}F_b/\mathrm{d}s$ at $s = V$, in N s/m. A decreasing friction curve has $F_b'(V) < 0$: the coefficient of velocity is negative.

Check the energy: why negative damping causes growth

The disturbance has energy $E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2$, measured about the shifted equilibrium. Differentiate each term:

$$\dot{E} = m\dot{x}\ddot{x} + kx\dot{x} = \dot{x}(m\ddot{x} + kx).$$

From the local equation, $m\ddot{x} + kx = -F_b'(V)\dot{x}$. Thus

$$\boxed{\dot{E} = -F_b'(V)\dot{x}^2 > 0 \quad \text{if } F_b'(V) < 0,\ \dot{x} \neq 0.}$$

What this calculation does and does not predict

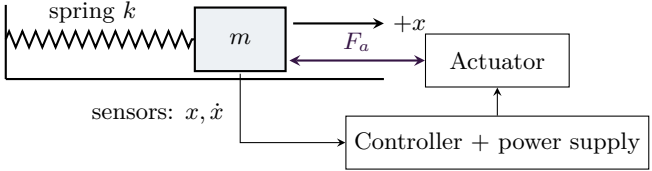
Small disturbances gain energy and grow. It does not yet predict a final amplitude: the local approximation cannot be extended to arbitrarily large motion.

A speed-independent friction magnitude has $F_b'(V) = 0$ and does not produce this local growth mechanism. Sticking and reversal of sliding require a different analysis.

We next use a specified feedback law to model both small-motion growth and large-motion saturation.

A powered cart makes the energy source explicit

Attach an actuator to a spring-mounted cart. Sensors measure displacement and velocity; a controller commands the actuator to assist or brake the motion.



x: displacement from spring equilibrium (m); F_a: actuator force (N). The same m (kg) and k (N/m) describe the cart and spring. Assume ideal force control without delay.

$$m\ddot{x} + kx = F_a.$$

The power supply provides energy during assistance. During braking the drive absorbs or dissipates mechanical energy. The command depends on the motion, not on a prescribed periodic clock.

Build a force law with assistance and braking

For the powered cart, Newton's equation is $m\ddot{x} + kx = F_a$. We now specify the actuator command.

First contribution: reinforce velocity

Choose $+c\dot{x}$, where $c > 0$ has units N s/m. It points with velocity, so its power is $c\dot{x}^2 > 0$ during motion.

Second contribution: make braking stronger away from the centre

Choose $-(c/L^2)x^2\dot{x}$, where $L > 0$ is a displacement scale (m). Its power is $-(c/L^2)x^2\dot{x}^2 \leq 0$.

$$F_a = c\dot{x} - \frac{c}{L^2}x^2\dot{x} = c\left(1 - \frac{x^2}{L^2}\right)\dot{x}.$$

At $|x| = L$, the two contributions cancel. This is a chosen feedback model, not a friction law derived from the belt example. Both illustrate motion-dependent energy transfer.

Follow force and velocity through the energy balance

For spring energy plus kinetic energy, $E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2$,

$$\dot{E} = (m\ddot{x} + kx)\dot{x} = F_a\dot{x} = c(1 - x^2/L^2)\dot{x}^2.$$

Inside $|x| < L$: assistance

The bracket is positive. If velocity is rightward, force is rightward; if velocity is leftward, force is leftward. In both directions $F_a\dot{x} > 0$.

Outside $|x| > L$: braking

The bracket is negative. Force opposes velocity in either direction, so $F_a\dot{x} < 0$.

At $|x| = L$, the command is zero. At a turning point, $\dot{x} = 0$, so force and instantaneous power are also zero. Position alone does not imply nonzero power.

Relate the active force to an effective damping coefficient

Write the actuator equation first with force on the right:

$$m\ddot{x} + kx = c(1 - x^2/L^2)\dot{x}.$$

Move the entire right-hand term to the left. Its sign reverses:

$$\boxed{m\ddot{x} + c(x^2/L^2 - 1)\dot{x} + kx = 0.}$$

Read the coefficient multiplying velocity

Near $x = 0$, it is approximately $-c$: negative damping supplies energy. For $|x| > L$, it is positive: damping removes energy.

A large oscillation passes through both regions on every cycle. It can still gain energy near the centre while losing more energy farther out.

The eventual amplitude is determined by the total input and loss over the cycle, not by setting the instantaneous coefficient to zero.

Worked example: actuator force and power

Take feedback strength $c = 1$ N s/m, threshold $L = 0.010$ m and rightward velocity $\dot{x} = 0.020$ m/s.

Inside the threshold: $x = 0.005$ **m**

$$1 - \frac{x^2}{L^2} = 1 - (0.5)^2 = 0.75, \qquad F_a = (1)(0.75)(0.020) = +0.015 \text{ N}.$$

$$F_a \dot{x} = (0.015)(0.020) = +0.00030 \text{ W}.$$

Outside the threshold: $x = 0.020$ **m**

$$1 - \frac{x^2}{L^2} = 1 - 2^2 = -3, \qquad F_a = (1)(-3)(0.020) = -0.060 \text{ N}.$$

$$F_a \dot{x} = (-0.060)(0.020) = -0.00120 \text{ W}.$$

Inside, the actuator pushes right and adds energy; outside, it pushes left and brakes the cart. Reversing velocity reverses the force too, so the power signs stay the same. At release from rest, both force and power are zero.

Why introduce dimensionless displacement and time?

The physical model contains a displacement threshold L and a natural frequency $\omega_0 = \sqrt{k/m}$. Use these scales to compare motions without carrying their units through every step:

$$u = \frac{x}{L}, \qquad \tau = \omega_0 t.$$

$u = 1$ means $x = L$, the sign-change threshold of the feedback. A leading linear cycle takes 2π units of τ. Both variables are dimensionless.

Primes now mean $\mathrm{d}/\mathrm{d}\tau$. Since $x = Lu(\tau)$, the chain rule gives

$$\dot{x} = L\frac{\mathrm{d}u}{\mathrm{d}\tau}\frac{\mathrm{d}\tau}{\mathrm{d}t} = L\omega_0 u', \qquad \ddot{x} = L\omega_0^2 u''.$$

We have changed units, not the physical model. The next substitution collects its damping and gain strength into one parameter.

Obtain the Van der Pol equation from the force law

Insert $x = Lu$, $\dot{x} = L\omega_0 u'$ and $\ddot{x} = L\omega_0^2 u''$ into Newton's equation:

$$mL\omega_0^2 u'' + kLu = c(1-u^2)L\omega_0 u'.$$

Divide every term by $mL\omega_0^2 = kL$:

$$u'' + u = \frac{c}{m\omega_0}(1-u^2)u'.$$

$$\boxed{u'' + u = \varepsilon(1-u^2)u', \qquad \varepsilon = \frac{c}{m\omega_0} > 0.}$$

$\varepsilon \ll 1$ means weak damping and gain over a fast cycle. It does not require $u \ll 1$. Unlike Week 3's conservative expansion, ε here does not count a small displacement amplitude.

Energy identifies where the oscillator gains and loses

Scale the physical energy using $x = Lu$, $\dot{x} = L\omega_0 u'$ and $m\omega_0^2 = k$:

$$E = \tfrac{1}{2}mL^2\omega_0^2 u'^2 + \tfrac{1}{2}kL^2u^2 = \tfrac{1}{2}kL^2(u'^2 + u^2).$$

Define dimensionless energy $\mathcal{E} = E/(kL^2)$. Differentiate and use the equation of motion:

$$\mathcal{E}' = u'(u'' + u) = \boxed{\varepsilon(1 - u^2)u'^2}.$$

$|u| < 1$ Energy enters while the system moves.

$|u| > 1$ Energy leaves while the system moves.

$u' = 0$ Instantaneous power is zero at a turning point.

This is an exact instantaneous balance. A sustained oscillation needs a balance over an entire cycle.

Does the sign-change position set the sustained amplitude?

Prediction: could the sustained amplitude be 1?

If the turning points were $u = \pm 1$, the moving part of the orbit would lie entirely in $|u| < 1$. Thus $\mathcal{E}' > 0$ between the turning points: there would be no loss interval to balance the input.

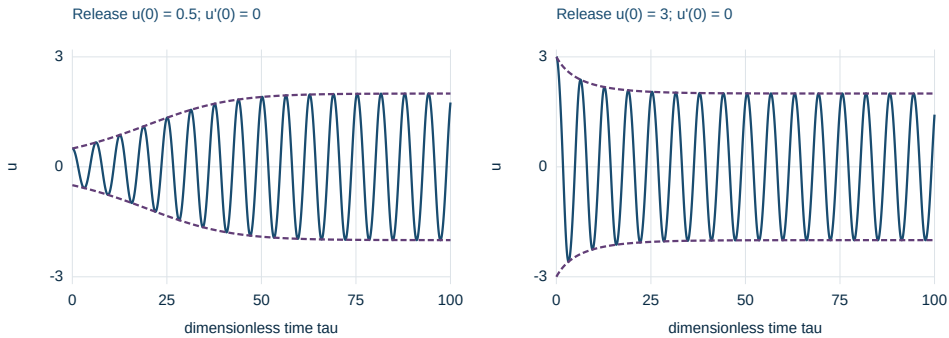
For periodic motion, the final energy must equal the initial energy. Integrate the energy rate over one dimensionless period T_τ:

$$\mathcal{E}(T_\tau) - \mathcal{E}(0) = \int_0^{T_\tau} \varepsilon(1 - u^2)u'^2 \, \mathrm{d}\tau = 0.$$

The threshold $|u| = 1$ locates the change in instantaneous power. It does not determine the orbit's amplitude. We must balance central energy input against outer energy loss.

This whole-cycle balance is the physical question behind **amplitude selection**.

Small and large releases approach sustained motion



$\varepsilon = 0.10$. Blue: direct integration. Purple dashed: the envelope derived next. Both releases approach the same amplitude, despite very different initial energies.

What are we trying to predict about the oscillation?

The previous time histories showed small releases growing and large releases shrinking toward similar peaks. The feedback model is

$$u'' + u = \varepsilon(1 - u^2)u', \qquad 0 < \varepsilon \ll 1.$$

$u = x/L$ is displacement scaled by the feedback threshold; $\tau = \omega_0 t$ is dimensionless time; primes mean $\mathrm{d}/\mathrm{d}\tau$.

Three questions about the observed motion

What peak size is approached? How quickly is it approached? Does the phase accumulate an additional drift?

Solving for every rapid oscillation is more information than these questions need. We seek equations for the slowly changing amplitude and phase.

First change coordinates exactly, so no motion is lost. Then average the rapid variation to obtain an approximate envelope equation. Keeping these two steps separate tells us where approximation enters.

Why describe motion by an amplitude and a phase?

Displacement alone does not tell us where the oscillator is in its cycle. At $u = 0$, it can be moving right or left. We need the state (u, u'), where primes mean $\mathrm{d}/\mathrm{d}\tau$.

Use the linear oscillator as a guide

For $u = a\cos\phi$ and $u' = -a\sin\phi$ with fixed amplitude,

$$u^2 + u'^2 = a^2(\cos^2\phi + \sin^2\phi) = a^2.$$

This suggests describing each nonzero state by its distance from the origin and its position around the circle:

$$a = \sqrt{u^2 + u'^2}, \qquad \cos\phi = \frac{u}{a}, \qquad \sin\phi = -\frac{u'}{a}.$$

These definitions also work for the nonlinear motion. Its radius a and angle ϕ may change nonuniformly. The minus sign makes phase increase in the direction of the linear motion.

The velocity relation is a coordinate definition

Our two definitions give, at each instant,

$$\boxed{u = a\cos\phi, \qquad u' = -a\sin\phi.}$$

Check three familiar states

(u, u')	a	ϕ on the first cycle
$(2, 0)$	2	0
$(0, -2)$	2	$\pi/2$
$(-2, 0)$	2	π

We have not differentiated $a\cos\phi$ while ignoring a'. We have defined a and ϕ from two state components. Differentiating must now produce a relation between their rates.

The exact radius can fluctuate within a cycle. Its slowly varying approximation gives the envelope; multiplying by L converts it to metres. At rest, $a = 0$ and phase is undefined.

Changing amplitude and phase must give the same velocity

Write the total phase as $\phi = \tau + \beta(\tau)$, where β is the phase offset in radians. Then $\phi' = 1 + \beta'$.

Differentiate $u = a\cos\phi$ with the product and chain rules:

$$u' = a'\cos\phi - a\sin\phi\,\phi' = a'\cos\phi - a(1+\beta')\sin\phi.$$

It must equal the velocity defined by the state coordinates, $u' = -a\sin\phi$:

$$a'\cos\phi - a\sin\phi - a\beta'\sin\phi = -a\sin\phi.$$

$$\boxed{a'\cos\phi - a\beta'\sin\phi = 0.}$$

We have not ignored a' or β'. This constraint links them. A second equation from Newton's law will determine both rates.

Derive the second amplitude–phase equation

Differentiate the defined velocity $u' = -a\sin\phi$, with $\phi' = 1 + \beta'$:

$$u'' = -a'\sin\phi - a(1+\beta')\cos\phi.$$

Insert this, $u = a\cos\phi$ and $u' = -a\sin\phi$ into $u'' + u = \varepsilon(1-u^2)u'$:

$$-a'\sin\phi - a(1+\beta')\cos\phi + a\cos\phi = -\varepsilon a(1 - a^2\cos^2\phi)\sin\phi.$$

The terms $-a\cos\phi$ and $+a\cos\phi$ cancel. Multiply what remains by -1:

$$\boxed{a'\sin\phi + a\beta'\cos\phi = \varepsilon a(1 - a^2\cos^2\phi)\sin\phi.}$$

Together with $a'\cos\phi - a\beta'\sin\phi = 0$, this determines both unknown rates. No averaging has been used.

Eliminate the phase rate to find the amplitude rate

We have two equations for a' and $a\beta'$:

$$a' \cos\phi - a\beta' \sin\phi = 0,$$
$$a' \sin\phi + a\beta' \cos\phi = \varepsilon a(1 - a^2 \cos^2\phi) \sin\phi.$$

Multiply the first by $\cos\phi$ and the second by $\sin\phi$, then add. The mixed terms cancel:

$$a'(\cos^2\phi + \sin^2\phi) + a\beta'(-\sin\phi\cos\phi + \cos\phi\sin\phi)$$
$$= \varepsilon a(1 - a^2\cos^2\phi)\sin^2\phi.$$

$$\boxed{a' = \varepsilon a(1 - a^2\cos^2\phi)\sin^2\phi.}$$

This is still an exact instantaneous rate. Its dependence on phase describes growth and loss within each cycle.

Eliminate the amplitude rate to find the phase rate

This time multiply the second equation by $\cos\phi$ and subtract the first multiplied by $\sin\phi$.

$$\begin{aligned}&[a'\sin\phi + a\beta'\cos\phi]\cos\phi\\ &\quad - [a'\cos\phi - a\beta'\sin\phi]\sin\phi\\ &= \varepsilon a(1 - a^2\cos^2\phi)\sin\phi\cos\phi.\end{aligned}$$

The a' terms cancel. The remaining left side is $a\beta'(\cos^2\phi + \sin^2\phi) = a\beta'$. Divide by $a > 0$:

$$\boxed{\beta' = \varepsilon(1 - a^2\cos^2\phi)\sin\phi\cos\phi.}$$

The total phase rate is $\phi' = 1 + \beta'$. Here 1 is the linear oscillator's phase rate; β' is the motion-dependent correction.

What does holding amplitude fixed for one cycle mean?

The exact rates contain a small factor ε. For amplitudes of order one,

$$a' = O(\varepsilon), \qquad \phi' = 1 + O(\varepsilon).$$

Follow one cycle starting at time τ_0

Phase advances by about 2π, but amplitude changes only by $O(\varepsilon)$. To estimate the change in that cycle, evaluate the rate using the current amplitude $a(\tau_0)$.

For the following cycle, use the updated amplitude. Freezing it inside one integral does not freeze it for the whole motion.

$$\text{change per cycle} = O(\varepsilon), \qquad \text{number of cycles for a large change} = O(1/\varepsilon).$$

This is why the motion has a fast oscillation and a slow envelope. The approximation requires weak within-cycle variation, not a constant envelope.

Why the freezing error is beyond first order

Write the amplitude equation briefly as $a' = \varepsilon F(a, \phi)$, where F denotes the already-derived trigonometric expression.

During one cycle, let the amplitude differ from its starting value by $\delta a = O(\varepsilon)$. Taylor expansion gives

$$\begin{aligned} \varepsilon F(a + \delta a, \phi) &= \varepsilon F(a, \phi) + \varepsilon \frac{\partial F}{\partial a} \delta a + \cdots \\ &= \varepsilon F(a, \phi) + O(\varepsilon^2). \end{aligned}$$

Why average over phase rather than time?

Since $\phi' = 1 + O(\varepsilon)$, $\mathrm{d}\tau/\mathrm{d}\phi = 1 + O(\varepsilon)$. This also changes an $O(\varepsilon)$ rate only at $O(\varepsilon^2)$.

First-order averaging keeps the $O(\varepsilon)$ accumulated rate. The discarded terms matter when higher accuracy is required.

Evaluate the two cycle averages explicitly

For a phase-dependent quantity q, define $\langle q \rangle = (2\pi)^{-1} \int_0^{2\pi} q \, \mathrm{d}\phi$, holding the current amplitude fixed.

First average: $\sin^2 \phi$

$$\langle \sin^2 \phi \rangle = \frac{1}{2\pi} \int_0^{2\pi} \frac{1 - \cos 2\phi}{2} \, \mathrm{d}\phi = \frac{1}{2\pi} \left[\frac{\phi}{2} - \frac{\sin 2\phi}{4} \right]_0^{2\pi} = \frac{1}{2}.$$

Second average: $\cos^2 \phi \sin^2 \phi$

$$\cos^2 \phi \sin^2 \phi = \tfrac{1}{4} \sin^2 2\phi = \tfrac{1}{8}(1 - \cos 4\phi),$$

$$\langle \cos^2 \phi \sin^2 \phi \rangle = \frac{1}{2\pi} \left[\frac{\phi}{8} - \frac{\sin 4\phi}{32} \right]_0^{2\pi} = \frac{1}{8}.$$

The oscillating sine terms return to their starting values. Only the constant parts contribute to the cycle averages.

Insert the averages to obtain the slow amplitude equation

Expand the exact amplitude rate before taking its average:

$$a' = \varepsilon a \sin^2\phi - \varepsilon a^3 \cos^2\phi \sin^2\phi.$$

Within the integral, εa and εa^3 are constant factors. Therefore

$$\langle a' \rangle = \varepsilon a \left(\frac{1}{2}\right) - \varepsilon a^3 \left(\frac{1}{8}\right).$$

Use this mean rate to evolve the envelope

$$\boxed{a' = \frac{\varepsilon a}{2} - \frac{\varepsilon a^3}{8}.}$$

From this point, a denotes the leading averaged envelope unless stated otherwise. The linear term promotes growth; the cubic term limits growth. The exact instantaneous radius still has small within-cycle variations.

How does a cycle average become an evolution equation?

Over one nearly harmonic cycle, the amplitude change is the integral of its rate:

$$\Delta a = \int_{\tau_0}^{\tau_0+2\pi} a'(\tau)\,\mathrm{d}\tau \approx 2\pi\,\frac{\varepsilon a}{2}(1 - a^2/4).$$

Divide by the cycle duration: the mean change per unit time is $\Delta a/(2\pi)$. Let the envelope evolve at this mean rate, updating a as it changes.

A small-change example

For $a = 1$ and $\varepsilon = 0.01$,

$$a'_{\text{averaged}} = \frac{0.01(1)}{2}(1 - 1/4) = 0.00375,$$

$$\Delta a \approx 2\pi(0.00375) = 0.02356.$$

The amplitude changes by about 2.36% in this cycle. The exact rate fluctuates throughout the cycle; the envelope retains the accumulated change. This estimate is not an exact cycle-by-cycle update formula.

Phase advances and delays cancel at first order

The instantaneous phase-offset rate is

$$\beta' = \varepsilon(\sin\phi\cos\phi - a^2\cos^3\phi\sin\phi).$$

Hold a fixed during the integral. The antiderivatives are $-\cos^2\phi/2$ and $-\cos^4\phi/4$, respectively:

$$\begin{aligned}\langle\beta'\rangle &= \frac{\varepsilon}{2\pi}\int_0^{2\pi}(\sin\phi\cos\phi - a^2\cos^3\phi\sin\phi)\,\mathrm{d}\phi \\ &= \frac{\varepsilon}{2\pi}\left[-\frac{1}{2}\cos^2\phi + \frac{a^2}{4}\cos^4\phi\right]_0^{2\pi} = 0.\end{aligned}$$

Cosine is 1 at both endpoints, so the endpoint values cancel. Instantaneous β' need not vanish; its advances and delays balance over one cycle at this order.

This means no first-order mean frequency correction. It does not mean the exact frequency remains unchanged at higher orders.

Amplitude selection: find the possible steady envelopes

Amplitude selection asks which sustained amplitude is reached after the transient. The release sets the starting amplitude, but energy exchange sets its eventual value.

$$a' = \frac{\varepsilon a}{2}\left(1 - \frac{a^2}{4}\right), \qquad \varepsilon > 0.$$

A steady envelope has $a' = 0$. A product is zero when at least one factor is zero:

$$a = 0 \qquad \text{or} \qquad 1 - a^2/4 = 0.$$

$$a^2 = 4 \quad \Rightarrow \quad a = 2 \quad \text{because amplitude is nonnegative.}$$

$a = 0$ is rest. $a = 2$ is a candidate sustained oscillation. Finding the candidates does not yet tell us whether nearby amplitudes move toward them.

Which steady amplitude does a nonzero release approach?

Read the sign of $a' = \varepsilon a(1 - a^2/4)/2$ for $a > 0$:

amplitude	$1 - a^2/4$	change
$0 < a < 2$	positive	$a' > 0$: grows
$a > 2$	negative	$a' < 0$: decreases

Both sides move toward 2. Exact rest stays at zero, but a small positive disturbance grows away from zero.

Translate back to displacement

$$u(\tau) \approx a(\tau)\cos(\tau + \beta_0), \qquad x(t) \approx La(\omega_0 t)\cos(\omega_0 t + \beta_0).$$

β_0 is the initial phase offset. The selected physical amplitude is approximately $2L$, with leading frequency ω_0. This is a first-order result, not an exact sinusoidal solution.

Use energy to understand why the selected amplitude is 2

The exact scaled energy rate is $\mathcal{E}' = \varepsilon(1-u^2)u'^2$. On a nearly harmonic cycle, insert $u \approx a\cos\phi$ and $u' \approx -a\sin\phi$:

$$\mathcal{E}' \approx \varepsilon a^2 \sin^2\phi - \varepsilon a^4 \cos^2\phi \sin^2\phi.$$

Average the two contributions using $\langle \sin^2\phi \rangle = 1/2$ and $\langle \cos^2\phi \sin^2\phi \rangle = 1/8$:

$$\langle \mathcal{E}' \rangle \approx \underbrace{\varepsilon a^2/2}_{\text{assistance}} - \underbrace{\varepsilon a^4/8}_{\text{braking}}.$$

Assistance scales as a^2, but braking scales as a^4. Braking therefore catches up as amplitude grows. Balance gives

$$\frac{a^2}{2} = \frac{a^4}{8} \quad \Rightarrow \quad a^2 = 4 \quad \Rightarrow \quad a = 2 \quad (a > 0).$$

The value 2 comes from a whole-cycle balance. The position $|u| = 1$ only marks where instantaneous power changes sign.

Check that energy balance gives the same envelope rate

For the exact radius $a = \sqrt{u^2 + u'^2}$, $\mathcal{E} = a^2/2$. For the slowly varying envelope, use this relation at leading order:

$$\mathcal{E}_{\text{slow}} \approx \tfrac{1}{2}a^2, \qquad \mathcal{E}'_{\text{slow}} \approx aa'.$$

Equate its rate to the averaged gain minus loss:

$$aa' = \varepsilon(a^2/2 - a^4/8).$$

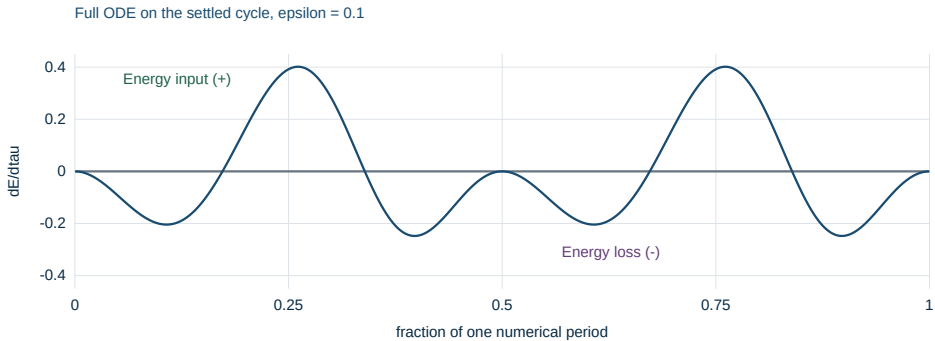
$$\boxed{a' = \varepsilon(a/2 - a^3/8) \quad (a > 0).}$$

At the selected amplitude

For $a = 2$, assistance is 2ε and braking is 2ε, so the net average is zero.

This independently recovers the envelope equation. It does not make instantaneous energy constant. Keep the separate rest solution $a = 0$, excluded temporarily by division.

Energy fluctuates even on the limiting cycle



The vertical axis is energy rate: above zero means gain, below zero means loss. Time runs left to right through one period. The signed areas cancel, so energy returns to its starting value; it need not stay constant within the cycle.

Solve for the envelope: first simplify the powers

We know where amplitude goes. Now ask how fast it gets there. For a positive release from rest, let $a(0) = a_0 = R/L$.

$$a' = \frac{\varepsilon a}{2}(1 - a^2/4).$$

Because the equation contains a and a^3, set $b = a^2$. This is a temporary algebraic variable. Differentiate:

$$b' = 2aa' = 2a\frac{\varepsilon a}{2}(1 - a^2/4) = \varepsilon b(1 - b/4).$$

For $b \neq 0, 4$, divide by the factors containing b and separate variables:

$$\frac{\mathrm{d}b}{b(1 - b/4)} = \varepsilon\,\mathrm{d}\tau.$$

The constant solutions $b = 0$ and $b = 4$ must be kept separately. Dividing by a factor cannot be used to prove its zero is not a solution.

Integrate the separated amplitude equation

The left side contains a rational expression. Rewrite it as two simple fractions:

$$\frac{1}{b(1-b/4)} = \frac{4}{b(4-b)} = \frac{1}{b} + \frac{1}{4-b}.$$

Check by adding the fractions: their numerator is $(4-b)+b=4$. Now integrate:

$$\int \frac{\mathrm{d}b}{b} + \int \frac{\mathrm{d}b}{4-b} = \ln b - \ln|4-b|.$$

The minus sign in the second integral comes from $\mathrm{d}(4-b) = -\mathrm{d}b$. Here $b>0$, so $\ln b$ is defined.

Using the initial value $b(0)=a_0^2$ gives the definite relation

$$\ln\left|\frac{b}{4-b}\right| - \ln\left|\frac{a_0^2}{4-a_0^2}\right| = \varepsilon\tau.$$

Isolate amplitude from the integrated equation

Exponentiate the integrated equation. A solution cannot cross the equilibrium $b = 4$, so the ratio keeps its initial sign:

$$\frac{b}{4-b} = \frac{a_0^2}{4-a_0^2} e^{\varepsilon\tau}.$$

Take reciprocals, expand the numerator, and solve for b:

$$\frac{4-b}{b} = \frac{4-a_0^2}{a_0^2} e^{-\varepsilon\tau},$$

$$\frac{4}{b} = 1 + \left(\frac{4}{a_0^2} - 1\right) e^{-\varepsilon\tau}, \qquad b = \frac{4}{1 + (4/a_0^2 - 1)e^{-\varepsilon\tau}}.$$

$$\boxed{a(\tau) = \frac{2}{\sqrt{1 + (4/a_0^2 - 1)e^{-\varepsilon\tau}}}.}$$

Check the envelope formula before using it

The solution for $a_0 > 0$ is

$$a(\tau) = \frac{2}{\sqrt{1 + (4/a_0^2 - 1)e^{-\varepsilon\tau}}}.$$

At release: $\tau = 0$

$$a(0) = \frac{2}{\sqrt{1 + (4/a_0^2 - 1)}} = \frac{2}{\sqrt{4/a_0^2}} = a_0.$$

After a long time: $\tau \to \infty$

The exponential tends to zero, so the denominator tends to 1 and $a \to 2$.

For $a_0 = 2$, the formula gives the constant solution 2. Exact rest $a_0 = 0$ must be retained separately because the calculation divided by amplitude. For every positive release, the denominator stays positive for forward time.

Worked example: calculate the envelope at $\tau = 20$

Take $\varepsilon = 0.10$ and release from rest at $u(0) = 0.50$, so $a_0 = 0.50$.

Evaluate the initial-amplitude factor

$$\frac{4}{a_0^2} - 1 = \frac{4}{(0.50)^2} - 1 = 16 - 1 = 15.$$

Evaluate the time factor and substitute

$$e^{-\varepsilon\tau} = e^{-0.10(20)} = e^{-2} \approx 0.135335,$$

$$a(20) = \frac{2}{\sqrt{1 + 15(0.135335)}} = \frac{2}{\sqrt{3.03003}} \approx 1.14896.$$

The envelope has grown from 0.50 to about 1.15. It is still below the selected value 2. This is an amplitude prediction; displacement at $\tau = 20$ also depends on phase.

Convert the envelope prediction to metres and seconds

Use mass $m = 1$ kg, stiffness $k = 100$ N/m, feedback strength $c = 1$ N s/m and threshold $L = 0.010$ m.

$$\omega_0 = \sqrt{k/m} = 10 \text{ rad/s}, \qquad \varepsilon = \frac{c}{m\omega_0} = \frac{1}{1(10)} = 0.10.$$

Since $\tau = \omega_0 t$ and $x = Lu$,

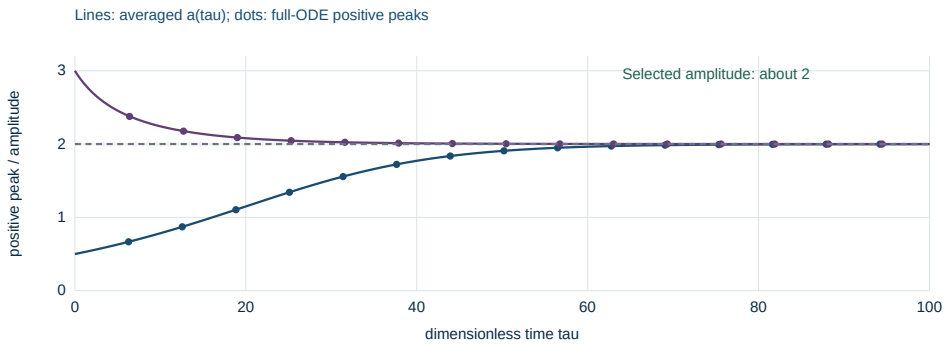
$$\tau = 20 \;\Rightarrow\; t = 20/10 = 2 \text{ s},$$

$$x_{\text{envelope}}(2 \text{ s}) \approx La(20) = (0.010)(1.14896) = 0.01149 \text{ m}.$$

$$R = La_0 = (0.010)(0.50) = 0.005 \text{ m}, \qquad x_{\text{limiting amplitude}} \approx 2L = 0.020 \text{ m}.$$

The initial release R is 5 mm, the predicted envelope after 2 s is 11.49 mm, and the limiting amplitude is approximately 20 mm. L is the feedback threshold, not the release.

Compare the predicted amplitude with numerical peaks



Both calculations use $\varepsilon = 0.10$ and releases from rest. Small discrepancies reflect the neglected waveform and higher-order corrections, not a different energy source.

What did averaging predict, and what remains approximate?

The envelope calculation answered the main questions for the standard weak Van der Pol oscillator:

$$a(\tau) = \frac{2}{\sqrt{1 + (4/a_0^2 - 1)e^{-\varepsilon\tau}}}, \qquad \beta(\tau) \approx \beta_0.$$

What we can reconstruct at leading order

$$u(\tau) \approx a(\tau)\cos(\tau + \beta_0).$$

The envelope approaches 2, and there is no first-order mean phase drift. The numerical peaks test the amplitude prediction.

Why use multiple scales next?

It provides a second route to the same slow evolution and also exposes the first waveform correction and the correction needed at release.

The changing envelope is the main result. A cosine multiplied by that envelope is an approximation, so its derivative need not satisfy the release velocity exactly without a correction.

Check the amplitude law using multiple scales

Multiple scales will recover the amplitude law by removing secular terms. Since $a'/a = O(\varepsilon)$, over an interval $\Delta\tau$,

$$\frac{\Delta a}{a} \sim \varepsilon \Delta\tau.$$

An appreciable amplitude change therefore needs $\Delta\tau \sim 1/\varepsilon$. Define fast and slow times

$$T_0 = \tau, \qquad T_1 = \varepsilon\tau, \qquad u = u_0(T_0, T_1) + \varepsilon u_1(T_0, T_1) + \cdots .$$

Here u_0 is the leading dimensionless displacement and u_1 its first correction. In Week 3, $x = \varepsilon x_1 + \cdots$ counted a small amplitude; here u_0 is order one and ε counts weak damping.

Both times are dimensionless here because the starting time is $\tau = \omega_0 t$. A comma denotes partial differentiation: $u_{0,T_1} = \partial u_0/\partial T_1$.

Apply the chain rule to a function of two times

Treat $T_0 = \tau$ and $T_1 = \varepsilon\tau$ as independent variables while differentiating. Afterwards evaluate the result along these relations.

$$\frac{\mathrm{d}T_0}{\mathrm{d}\tau} = 1, \qquad \frac{\mathrm{d}T_1}{\mathrm{d}\tau} = \varepsilon.$$

For $u = u_0(T_0, T_1) + \varepsilon u_1(T_0, T_1) + \cdots$, differentiate each term:

$$\frac{\mathrm{d}u_0}{\mathrm{d}\tau} = u_{0,T_0} + \varepsilon u_{0,T_1}, \qquad \varepsilon\frac{\mathrm{d}u_1}{\mathrm{d}\tau} = \varepsilon u_{1,T_0} + \varepsilon^2 u_{1,T_1}.$$

Keep terms through first order:

$$\boxed{u' = u_{0,T_0} + \varepsilon(u_{0,T_1} + u_{1,T_0}) + O(\varepsilon^2).}$$

A comma denotes a partial derivative. The slow derivative is multiplied by ε because T_1 changes ε times as fast as τ.

Differentiate again: where does the factor 2 come from?

Start from $u' = u_{0,T_0} + \varepsilon u_{0,T_1} + \varepsilon u_{1,T_0} + O(\varepsilon^2)$.

Differentiate each retained contribution

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}\tau} u_{0,T_0} &= u_{0,T_0T_0} + \varepsilon u_{0,T_0T_1}, \\
\frac{\mathrm{d}}{\mathrm{d}\tau} (\varepsilon u_{0,T_1}) &= \varepsilon u_{0,T_1T_0} + O(\varepsilon^2), \\
\frac{\mathrm{d}}{\mathrm{d}\tau} (\varepsilon u_{1,T_0}) &= \varepsilon u_{1,T_0T_0} + O(\varepsilon^2).
\end{aligned}$$

For smooth functions, the two mixed derivatives are equal. Adding gives

$$\boxed{u'' = u_{0,T_0T_0} + \varepsilon(2u_{0,T_0T_1} + u_{1,T_0T_0}) + O(\varepsilon^2).}$$

One mixed derivative comes from differentiating the fast term slowly; the other comes from differentiating the slow term quickly. Both contribute to acceleration.

Match equal powers of the damping parameter

Insert the derivative expansion into $u'' + u = \varepsilon(1 - u^2)u'$. The right side already has a factor ε, so only its leading factors are needed:

$$\varepsilon(1 - u^2)u' = \varepsilon(1 - u_0^2)u_{0,T_0} + O(\varepsilon^2).$$

Before collecting orders, the complete substitution is

$$\begin{aligned} u_{0,T_0T_0} + u_0 + \varepsilon(u_{1,T_0T_0} + u_1 + 2u_{0,T_0T_1}) \\ = \varepsilon(1 - u_0^2)u_{0,T_0} + O(\varepsilon^2). \end{aligned}$$

The coefficients of each retained power must agree:

$$\begin{aligned} O(1): \quad & u_{0,T_0T_0} + u_0 = 0, \\ O(\varepsilon): \quad & u_{1,T_0T_0} + u_1 = -2u_{0,T_0T_1} + (1 - u_0^2)u_{0,T_0}. \end{aligned}$$

Solve the leading oscillator before evaluating the correction

At leading order, $u_{0,T_0T_0} + u_0 = 0$. With T_1 held fixed, this is the familiar linear oscillator. Write its solution as

$$u_0 = a(T_1)\cos\phi, \qquad \phi = T_0 + \beta(T_1).$$

Fast differentiation holds a and β fixed. Slow differentiation must differentiate both:

$$u_{0,T_0} = -a\sin\phi,$$
$$u_{0,T_0T_1} = -a_{T_1}\sin\phi - a\beta_{T_1}\cos\phi.$$

The nonlinear contribution is

$$(1-u_0^2)u_{0,T_0} = (1-a^2\cos^2\phi)(-a\sin\phi) = -a\sin\phi + a^3\cos^2\phi\sin\phi.$$

The slow derivative supplies terms that can cancel the resonant part of the nonlinear contribution. We next identify that part.

Separate the fundamental and third harmonics

Use the double-angle and product-to-sum identities:

$$\cos^2\phi\sin\phi = \tfrac{1}{2}\sin\phi + \tfrac{1}{2}\sin\phi\cos 2\phi,$$

$$\sin\phi\cos 2\phi = \tfrac{1}{2}[\sin 3\phi + \sin(-\phi)] = \tfrac{1}{2}(\sin 3\phi - \sin\phi).$$

$$\boxed{\cos^2\phi\sin\phi = \tfrac{1}{4}(\sin\phi + \sin 3\phi).}$$

Insert this and the slow derivative into the first-order equation:

$$u_{1,T_0T_0} + u_1 = (2a_{T_1} - a + a^3/4)\sin\phi + 2a\beta_{T_1}\cos\phi + (a^3/4)\sin 3\phi.$$

The sine and cosine at frequency 1 resonate with the correction equation. The frequency-3 term produces a bounded third harmonic.

Why a resonant term makes the correction grow

Consider $u_{1,T_0T_0} + u_1 = \cos T_0$. A trial proportional to $\cos T_0$ is already a homogeneous solution; its left side is zero. Include a factor T_0 instead.

Try $u_1 = BT_0 \sin T_0$, with B constant in fast time:

$$\begin{aligned} u_{1,T_0} &= B \sin T_0 + BT_0 \cos T_0, \\ u_{1,T_0T_0} &= 2B \cos T_0 - BT_0 \sin T_0, \\ u_{1,T_0T_0} + u_1 &= 2B \cos T_0. \end{aligned}$$

Matching the right side gives $2B = 1$, hence $u_1 = \frac{1}{2} T_0 \sin T_0$. A resonant sine similarly gives $-\frac{1}{2} T_0 \cos T_0$. These are secular corrections: εu_1 becomes order one when T_0 is order $1/\varepsilon$. The small-correction expansion then fails.

Let the slow amplitude absorb the resonant growth

Multiple scales allows $a(T_1)$ and $\beta(T_1)$ to evolve. Choose their rates so that the growing correction is unnecessary:

$$2a_{T_1} - a + a^3/4 = 0, \qquad 2a\beta_{T_1} = 0.$$

Solve the first equation and divide the second by $2a$ for $a > 0$:

$$a_{T_1} = \frac{1}{2}(a - a^3/4), \qquad \beta_{T_1} = 0.$$

Return to the time variable used in averaging

Since $T_1 = \varepsilon\tau$,

$$\frac{\mathrm{d}a}{\mathrm{d}\tau} = \frac{\mathrm{d}a}{\mathrm{d}T_1}\frac{\mathrm{d}T_1}{\mathrm{d}\tau} = \varepsilon a_{T_1}.$$

$$\boxed{a' = \frac{\varepsilon a}{2}(1 - a^2/4), \qquad \beta' = 0.}$$

This is the same envelope equation found by averaging. The secular change has become slow evolution of the leading motion rather than growth of a small correction.

Calculate the bounded third-harmonic correction

After resonance removal, hold T_1 fixed and solve

$$u_{1,T_0T_0} + u_1 = \frac{a^3}{4}\sin 3\phi.$$

Try $u_1 = B\sin 3\phi$ for a particular solution. Since $\phi_{T_0} = 1$,

$$u_{1,T_0} = 3B\cos 3\phi, \qquad u_{1,T_0T_0} = -9B\sin 3\phi.$$
$$(-9+1)B\sin 3\phi = (a^3/4)\sin 3\phi \quad \Rightarrow \quad B = -a^3/32.$$

The full correction also contains the homogeneous solutions:

$$\boxed{u_1 = -\frac{a^3}{32}\sin 3\phi + C\cos\phi + D\sin\phi.}$$

C and D are coefficients that may depend on T_1. Initial conditions determine their initial values. Higher-order equations would determine their later slow evolution.

Use the release displacement to fix the first correction

Let $a_0 = R/L > 0$, with $u(0) = a_0$ and $u'(0) = 0$. Choose $a(0) = a_0$ and $\beta(0) = 0$, so $\phi(0) = 0$.

At release, evaluate the correction

$$u_1(0) = -\frac{a_0^3}{32}\underbrace{\sin 0}_{0} + C(0)\underbrace{\cos 0}_{1} + D(0)\underbrace{\sin 0}_{0} = C(0).$$

$$u(0) = u_0(0) + \varepsilon u_1(0) + \cdots = a_0 + \varepsilon C(0) + \cdots .$$

To preserve the specified release displacement through first order,

$$\boxed{C(0) = 0.}$$

The leading cosine already has the correct initial displacement. The correction must not add an extra first-order displacement. Initial velocity requires a separate check.

Use release from rest to fix the sine correction

The total velocity includes the slow derivative:

$$u' = u_{0,T_0} + \varepsilon(u_{0,T_1} + u_{1,T_0}) + \cdots .$$

At $\phi = 0$, $u_{0,T_0}(0) = 0$ and $u_{0,T_1}(0) = a_{T_1}(0)$. Differentiating the correction gives

$$u_{1,T_0} = -\frac{3a^3}{32}\cos 3\phi - C\sin\phi + D\cos\phi,$$

$$u'(0) = \varepsilon\left[a_{T_1}(0) - \frac{3a_0^3}{32} + D(0)\right] + \cdots .$$

Set the bracket to zero to satisfy release from rest:

$$\boxed{D(0) = \frac{3a_0^3}{32} - a_{T_1}(0), \qquad a_{T_1}(0) = \frac{a_0}{2}(1 - a_0^2/4).}$$

An envelope times a cosine generally has a small nonzero initial velocity when the envelope is growing. The sine correction cancels that mismatch through the retained order.

Check the initial correction numerically

Use the earlier release $a_0 = 0.50$ and $\varepsilon = 0.10$. The slow amplitude derivative at release is

$$a_{T_1}(0) = \frac{0.50}{2}\left(1 - \frac{0.50^2}{4}\right) = 0.25(0.9375) = 0.234375.$$

The third-harmonic velocity contribution and sine coefficient are

$$\frac{3a_0^3}{32} = \frac{3(0.125)}{32} = 0.01171875,$$

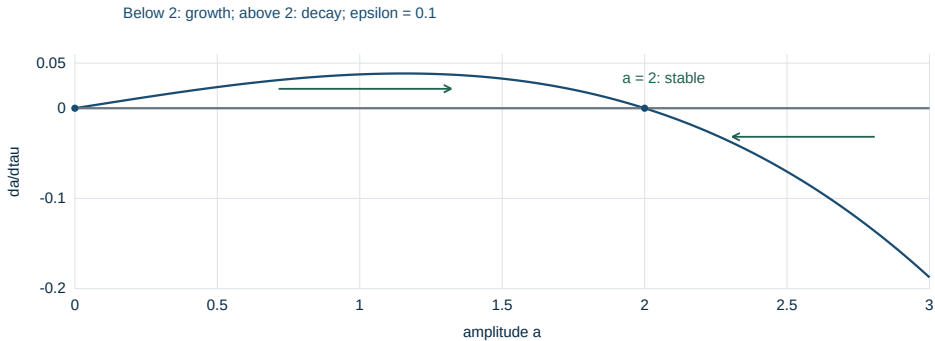
$$C(0) = 0, \qquad D(0) = 0.01171875 - 0.234375 = -0.22265625.$$

Check the total first-order initial velocity:

$$u'(0) = 0.10[0.234375 - 0.01171875 - 0.22265625] + O(\varepsilon^2) = O(\varepsilon^2).$$

The three contributions cancel. This checks the release conditions through first order; it does not turn the truncated expansion into an exact solution.

A stable amplitude is an equilibrium of the slow flow



The horizontal axis is amplitude; the vertical axis is its slow rate. Above zero, amplitude moves right; below zero, it moves left. Both arrows point toward 2. At exactly zero the oscillator stays at rest, but a small positive disturbance grows.

Turn an amplitude disturbance into a stability equation

Write the slow equation as $a' = f(a)$, where $f(a) = \varepsilon a/2 - \varepsilon a^3/8$. Let a_* be a steady amplitude, so $f(a_*) = 0$.

Introduce a small signed error $\eta = a - a_*$. Because a_* is constant, $a' = \eta'$. Taylor-expand the right side:

$$\eta' = f(a_* + \eta) = f(a_*) + f'(a_*)\eta + O(\eta^2).$$

$$\boxed{\eta' \approx f'(a_*)\eta.}$$

Here $f'(a_*)$ means the slope with respect to amplitude; η' means differentiation with respect to time τ. A positive or negative error represents starting above or below the steady amplitude.

Why a negative slope means recovery

In the linearized error equation, the slope $f'(a_*)$ is constant:

$$\eta' = f'(a_*)\eta.$$

For a nonzero error, separate and integrate from the initial time:

$$\int_{\eta(0)}^{\eta(\tau)} \frac{\mathrm{d}\eta}{\eta} = f'(a_*) \int_0^\tau \mathrm{d}s, \qquad \ln\left|\frac{\eta(\tau)}{\eta(0)}\right| = f'(a_*)\tau.$$

$$\boxed{\eta(\tau) \approx \eta(0) e^{f'(a_*)\tau}.}$$

Read the exponential

Negative slope: the magnitude of either signed error shrinks. Positive slope: it grows. Zero slope: this linear term gives no answer; retain nonlinear terms.

The exactly zero error stays zero. Stability concerns what happens after a small nonzero disturbance.

Apply the stability test to rest and sustained motion

Differentiate the averaged rate:

$$f'(a) = \frac{\varepsilon}{2} - \frac{3\varepsilon a^2}{8} = \frac{\varepsilon}{2}\left(1 - \frac{3a^2}{4}\right).$$

Near rest: $a_* = 0$

$$f'(0) = \varepsilon/2 > 0, \qquad \eta \approx \eta(0)e^{\varepsilon\tau/2}.$$

Near the selected amplitude: $a_* = 2$

$$f'(2) = \tfrac{\varepsilon}{2}(1-3) = -\varepsilon, \qquad \eta \approx \eta(0)e^{-\varepsilon\tau}.$$

For $\varepsilon = 0.10$, an error near the cycle falls to $1/e$ of its initial size after $\tau = 1/0.10 = 10$. This is its **e-folding recovery time**.

Exact rest remains a solution, but nonzero small disturbances grow. Near the sustained cycle, amplitude errors decay. Neither statement fixes the phase along the orbit.

A numerical disturbance: does the amplitude return?

For $\varepsilon = 0.10$, the steady amplitude is approximately 2 and a small error obeys $\eta(\tau) \approx \eta(0)e^{-0.10\tau}$.

Start slightly above the selected amplitude

If $a(0) = 2.10$, then $\eta(0) = 2.10 - 2 = 0.10$. After $\tau = 10$,

$$\eta(10) \approx 0.10e^{-1} = 0.0368, \qquad a(10) \approx 2.0368.$$

Start slightly below it

If $a(0) = 1.90$, then $\eta(0) = -0.10$, giving $\eta(10) \approx -0.0368$ and $a(10) \approx 1.9632$.

Both errors shrink in magnitude. The amplitude has not fully recovered after one e-folding time; about 36.8% of the initial error remains. These estimates linearize the slow flow near 2.

Constant amplitude does not mean a stationary mass

Introduce dimensionless velocity $v = u'$. The full state evolves according to

$$u' = v, \qquad v' = -u + \varepsilon(1 - u^2)v.$$

The equilibrium state is $(u, v) = (0, 0)$. A constant nonzero amplitude instead gives continuing motion. At leading order,

$$u \approx 2\cos(\tau + \beta_0), \qquad v \approx -2\sin(\tau + \beta_0).$$

Square and add to remove the time-dependent angle:

$$u^2 + v^2 \approx 4(\cos^2 + \sin^2) = 4.$$

The amplitude equilibrium $a = 2$ corresponds approximately to a circle of radius 2 in the displacement–velocity plane. The point travels around the circle; it does not stop.

What makes this periodic orbit a limit cycle?

A **limit cycle** is an isolated periodic orbit. Nearby motions may approach it even though they started with different energies.

Compare with a conservative linear oscillator

Every initial amplitude gives a different periodic circle. There is a whole family of circles, so none is isolated or selected by attraction.

For the weak Van der Pol oscillator

The amplitude equation sends nearby smaller and larger amplitudes toward the same nonzero value. This corresponds to attraction toward one periodic orbit.

Stability here means return toward the orbit. A phase shift only moves a point along that orbit. An autonomous system has no external clock that forces all trajectories to arrive at the same point at the same time.

This is **orbital stability**: amplitude errors decay, while a phase offset can remain.

Returning to the same orbit does not fix the phase

At leading order, consider two sustained motions with a quarter-cycle phase difference:

$$u_A = 2\cos\tau, \qquad v_A = -2\sin\tau,$$
$$u_B = 2\cos(\tau + \pi/2), \quad v_B = -2\sin(\tau + \pi/2).$$

Both satisfy $u^2 + v^2 = 4$. At $\tau = 0$, however, their states are different:

$$(u_A, v_A) = (2, 0), \qquad (u_B, v_B) = (0, -2).$$

What returns after a disturbance?

A small amplitude error decays toward the periodic orbit. A shift along that orbit need not disappear: the autonomous equation has no externally imposed phase reference.

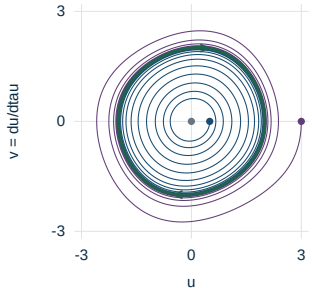
Orbital stability means attraction to the same geometric cycle, allowing a phase shift. It does not mean all displacement records become identical at the same time.

Phase portrait: attraction to an isolated cycle

Two releases approach the same isolated periodic orbit

Blue: release 0.5

Purple: release 3



Green: limiting cycle

Origin: unstable

Travel: clockwise

$\varepsilon = 0.10$. These are displacement–velocity trajectories, not displacement–time curves. The cycle is slightly noncircular; the circle of radius 2 is the leading weak-damping approximation.

Check departure from rest in the original equation

Near rest, both u and u' are small. Their product u^2u' is third order in the small disturbance, so the linear approximation is

$$u'' - \varepsilon u' + u = 0.$$

Try $u = e^{\lambda\tau}$. Then $u' = \lambda e^{\lambda\tau}$ and $u'' = \lambda^2 e^{\lambda\tau}$. Substitute and divide by the nonzero exponential:

$$\lambda^2 - \varepsilon\lambda + 1 = 0.$$

$$\lambda = \frac{\varepsilon \pm \sqrt{\varepsilon^2 - 4}}{2} = \frac{\varepsilon}{2} \pm i\sqrt{1 - \frac{\varepsilon^2}{4}} \qquad (0 < \varepsilon < 2).$$

λ is a dimensionless exponential growth rate. Its real part $\varepsilon/2 > 0$ multiplies a growing envelope; its imaginary part produces oscillation. Thus small disturbances spiral away from rest, consistent with averaging.

Vary the gain while keeping the saturation mechanism

In the standard force law $F_a = c(1-x^2/L^2)\dot{x}$, the coefficient 1 fixes the small-motion assistance. Replace it by an adjustable dimensionless gain μ:

$$F_a = c(\mu - x^2/L^2)\dot{x}, \qquad \boxed{u'' + u = \varepsilon(\mu - u^2)u'.}$$

What does turning this gain change?

For $\mu < 0$, both terms remove energy. For $\mu > 0$, motion near the centre gains energy while sufficiently large displacement causes braking. The standard model is $\mu = 1$.

$$\mathcal{E}' = \varepsilon(\mu - u^2)u'^2.$$

Keep $\varepsilon > 0$ fixed and small. At $\mu = 0$, nonlinear damping remains. We ask whether a stable oscillation appears with arbitrarily small amplitude as μ becomes positive.

Changing gain also changes the assistance region

The gain-controlled energy rate is

$$\mathcal{E}' = \varepsilon(\mu - u^2)u'^2.$$

For $\mu > 0$, power changes sign where $\mu - u^2 = 0$. Thus

$$|u| = \sqrt{\mu}, \qquad |x| = L\sqrt{\mu}.$$

Inside this region, motion gains energy; outside it, motion loses energy. The scale L is still fixed, but it equals the sign-change position only when $\mu = 1$.

Example: $\mu = 0.25$**,** $L = 0.010$ **m**

The sign-change position is $L\sqrt{\mu} = (0.010)(0.5) = 0.005$ m, or 5 mm.

This is not yet a predicted oscillation amplitude. We still need the cycle balance. For $\mu \leq 0$, there is no finite region of positive power.

Linearization locates where rest loses stability

For $u'' + u = \varepsilon(\mu - u^2)u'$, neglect the cubic term near rest:

$$u'' - \varepsilon\mu u' + u = 0.$$

Try a disturbance $u \propto e^{\lambda\tau}$, where λ is its dimensionless growth rate. Substitution gives

$$\lambda^2 - \varepsilon\mu\lambda + 1 = 0.$$

The quadratic formula gives, near $\mu = 0$,

$$\boxed{\lambda = \frac{\varepsilon\mu}{2} \pm i\sqrt{1 - \frac{\varepsilon^2\mu^2}{4}}.}$$

The real part determines growth or decay. The imaginary part gives the oscillation frequency of a small disturbance. We now track what changes as μ passes through zero.

Read the complex growth rate as a real motion

For $|\varepsilon\mu| < 2$, the linearized equation has a real solution

$$u(\tau) = e^{\varepsilon\mu\tau/2}\left[C\cos\left(\sqrt{1-\varepsilon^2\mu^2/4}\,\tau\right) + D\sin\left(\sqrt{1-\varepsilon^2\mu^2/4}\,\tau\right)\right].$$

C, D are dimensionless constants fixed by the initial displacement and velocity in this linearized problem.

The exponential answers the stability question

For $\mu < 0$ it shrinks; for $\mu > 0$ it grows. The sine and cosine show that the disturbance oscillates while its size changes.

At $\mu = 0$

The linear exponential becomes 1, but the linear model omits nonlinear damping. Zero linear decay does not prove that the full nonlinear motion persists forever.

Linearization tells us when rest loses stability. It cannot determine the amplitude that stops growth; that requires the nonlinear terms.

At the threshold, growth changes sign but oscillation remains

For fixed $\varepsilon > 0$, the real part is $\operatorname{Re}\lambda = \varepsilon\mu/2$:

$\mu < 0$	$\operatorname{Re}\lambda < 0$	small disturbances decay
$\mu = 0$	$\lambda = \pm i$	zero linear growth rate
$\mu > 0$	$\operatorname{Re}\lambda > 0$	small disturbances grow

The pair crosses the imaginary axis rather than just touching it, because

$$\frac{\mathrm{d}\operatorname{Re}\lambda}{\mathrm{d}\mu} = \frac{\varepsilon}{2} \neq 0.$$

At the crossing the frequency is nonzero. Linearization identifies the stability change, but cannot determine the amplitude or stability of a new cycle. The nonlinear amplitude equation supplies that information.

Average the gain-controlled model

Use the same amplitude–phase coordinates. The change is the constant gain μ in place of 1:

$$a' = \varepsilon a(\mu - a^2\cos^2\phi)\sin^2\phi.$$

Expand and use the already-evaluated averages 1/2 and 1/8:

$$\langle a' \rangle = \varepsilon\mu a\langle \sin^2\phi\rangle - \varepsilon a^3\langle \cos^2\phi \sin^2\phi\rangle,$$

$$\boxed{a' = \frac{\varepsilon\mu}{2}a - \frac{\varepsilon}{8}a^3 = \frac{\varepsilon a}{2}(\mu - a^2/4).}$$

Only the linear gain term changes with μ. The negative cubic term still limits growth. The first-order mean phase rate is zero by the same endpoint cancellation as before.

Find and test the new steady amplitude

Set $a' = \varepsilon a(\mu - a^2/4)/2 = 0$. Rest $a = 0$ always exists. For a nonzero amplitude,

$$\mu - a^2/4 = 0 \quad \Rightarrow \quad a^2 = 4\mu \quad \Rightarrow \quad \boxed{a_* = 2\sqrt{\mu} \quad (\mu > 0).}$$

There is no positive real branch for $\mu \leq 0$. To test the branch for $\mu > 0$, differentiate the averaged rate:

$$f'(a) = \frac{\varepsilon\mu}{2} - \frac{3\varepsilon a^2}{8}.$$

$$f'(a_*) = \frac{\varepsilon\mu}{2} - \frac{3\varepsilon(4\mu)}{8} = -\varepsilon\mu < 0.$$

The branch is attracting. As μ decreases toward zero, its amplitude tends continuously to zero and its recovery rate $\varepsilon\mu$ becomes slower.

At the threshold, retain the nonlinear damping

At $\mu = 0$, the averaged equation becomes $a' = -\varepsilon a^3/8$. Every positive amplitude still has a negative rate, even though the linear rate is zero.

For $a(0) > 0$, separate variables and integrate:

$$\int_{a(0)}^{a(\tau)} a^{-3}\,\mathrm{d}a = -\frac{\varepsilon}{8}\int_0^{\tau} \mathrm{d}s,$$

$$-\frac{1}{2a(\tau)^2} + \frac{1}{2a(0)^2} = -\frac{\varepsilon}{8}\tau.$$

Multiply by -2 and move the initial-value term to the right:

$$\frac{1}{a(\tau)^2} = \frac{1}{a(0)^2} + \frac{\varepsilon}{4}\tau.$$

$$\boxed{a(\tau) = \frac{1}{\sqrt{a(0)^{-2} + \varepsilon\tau/4}}.}$$

Why the threshold trajectory can look like a persistent ring

The threshold envelope is

$$a(\tau) = \frac{1}{\sqrt{a(0)^{-2} + \varepsilon\tau/4}}.$$

For large τ, the term proportional to τ dominates. Amplitude therefore decreases in proportion to $\tau^{-1/2}$, rather than exponentially.

Use the release shown in the phase portraits

With $a(0) = 1.4$, $\varepsilon = 0.10$ and $\tau = 180$,

$$a(180) \approx \frac{1}{\sqrt{1/1.4^2 + 0.10(180)/4}} = \frac{1}{\sqrt{0.5102 + 4.5}} \approx 0.447.$$

A visible radius remains at the end of this finite record, but it continues to decrease. A slowly shrinking spiral is not a limit cycle. The value here is the averaged prediction, not an exact numerical peak.

What makes this onset a supercritical Hopf bifurcation?

Put the linear and nonlinear results together near $\mu = 0$:

Before the threshold: $\mu < 0$

Small disturbances decay to rest. There is no nearby nonzero steady amplitude.

At the threshold: $\mu = 0$

The complex growth rates are $\lambda = \pm i$: the linear decay rate vanishes while oscillation frequency remains nonzero.

After the threshold: $\mu > 0$

Rest is unstable and a stable periodic motion appears with amplitude $a_* = 2\sqrt{\mu}$.

This local emergence of a stable cycle on the unstable-rest side is a **supercritical Hopf bifurcation**. Linearization locates the crossing; the negative cubic amplitude term establishes the branch's stability.

How would a subcritical onset differ?

For comparison only, suppose the cubic amplitude term had the opposite sign:

$$a' = \frac{\varepsilon\mu}{2}a + \frac{\varepsilon}{8}a^3.$$

A nonzero steady amplitude would satisfy $\mu + a^2/4 = 0$, giving

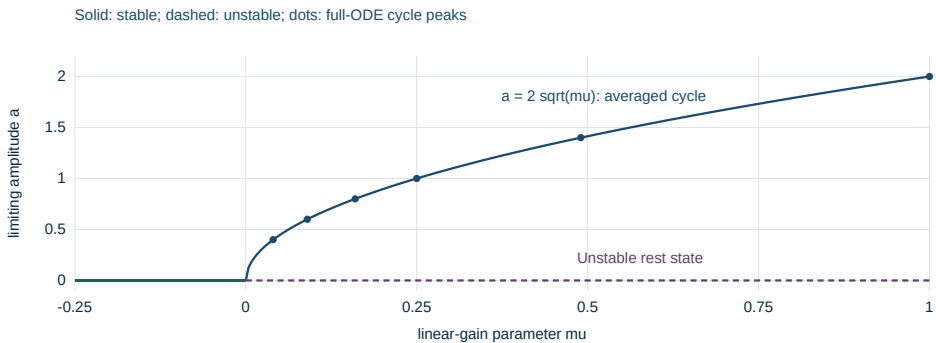
$$a_* = 2\sqrt{-\mu} \qquad (\mu < 0).$$

Its slope would be

$$f'(a_*) = \frac{\varepsilon\mu}{2} + \frac{3\varepsilon(-4\mu)}{8} = -\varepsilon\mu > 0.$$

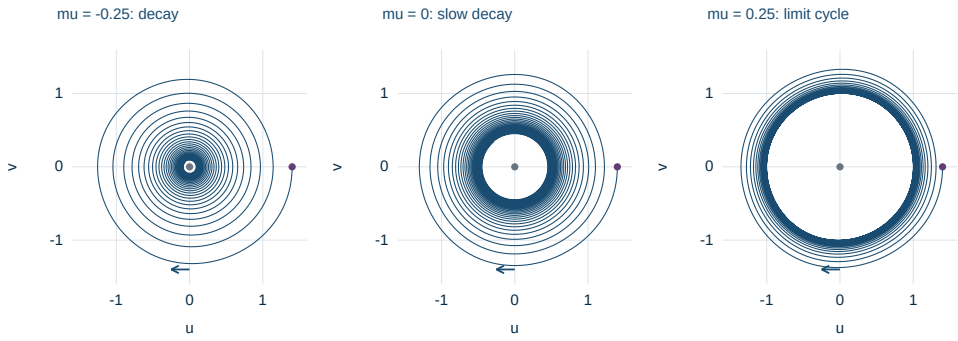
This nearby cycle is unstable and lies on the stable-rest side: the local subcritical case. This comparison does not change our Van der Pol model, whose cubic amplitude term is negative. Higher-order terms would be needed to predict any large stable cycle in the comparison model.

The cycle branch emerges from the rest state



At each horizontal value of gain μ, read the sustained amplitude vertically. Solid branches are stable; dashed rest is unstable; dots are numerical peaks. The branch begins at zero amplitude at $\mu = 0$. This graph shows long-time states, not a time history.

Phase portraits across the threshold



All three trajectories start at $(u, v) = (1.4, 0)$ and run to $\tau = 180$, with $\varepsilon = 0.10$. At the threshold, decay is algebraic and remains slow; a finite-time spiral is not evidence of a limit cycle.

Why varying the standard damping parameter is different

In the standard equation $u'' + u = \varepsilon(1 - u^2)u'$, the averaged nonzero amplitude is $a_* = 2$ for every small positive ε.

As $\varepsilon \to 0$, the selected amplitude does not shrink to zero. At exactly $\varepsilon = 0$, all amplitudes are periodic because the system becomes the linear conservative oscillator.

Thus varying ε through zero in the standard normalization is not the generic small-cycle onset just shown. For that onset, vary μ at fixed $\varepsilon > 0$, so the nonlinear saturation remains present at the threshold.

Worked Hopf example: predict the physical amplitude

Take $\varepsilon = 0.10$ and gain $\mu = 0.25$. Use mass $m = 1$ kg, stiffness $k = 100$ N/m and fixed displacement scale $L = 0.010$ m.

$$\omega_0 = \sqrt{100/1} = 10 \text{ rad/s}.$$

Solve the nonzero amplitude condition:

$$a_*^2 = 4\mu = 4(0.25) = 1 \quad \Rightarrow \quad a_* = 1.$$

$$\boxed{x_{\text{amplitude}} \approx L a_* = (0.010)(1) = 0.010 \text{ m}.}$$

The selected amplitude is approximately 10 mm, whereas the power sign changes at $L\sqrt{\mu} = 5$ mm. The orbit must enter the loss region to balance the input. Reducing gain from 1 to 0.25 reduces amplitude while nonlinear braking remains unchanged.

Worked Hopf example: distinguish recovery time from period

At $\varepsilon = 0.10$, $\mu = 0.25$, the slope at the selected amplitude is

$$f'(a_*) = -\varepsilon\mu = -(0.10)(0.25) = -0.025.$$

A small amplitude error therefore obeys $\eta' = -0.025\eta$, so

$$\eta(\tau) = \eta(0)e^{-0.025\tau}.$$

Time for the error to fall to $1/e$ of its initial size

$$0.025\tau = 1 \;\Rightarrow\; \tau = 40, \qquad t = \tau/\omega_0 = 40/10 = 4 \text{ s}.$$

Time for one oscillation

$$T \approx \frac{2\pi}{\omega_0} = \frac{2\pi}{10} = 0.6283 \text{ s}.$$

Recovery takes about $4/0.6283 = 6.4$ cycles. The mass keeps oscillating while its amplitude error gradually decays.

Check whether the envelope changes little in one cycle

First-order averaging assumes a nearly harmonic fast motion and slow amplitude evolution. In the standard model,

$$\frac{a'}{a} = \frac{\varepsilon}{2}(1 - a^2/4).$$

A fast cycle lasts about 2π. Multiplying rate by cycle duration estimates the fractional envelope change:

$$\left|\frac{\Delta a}{a}\right| \approx \left|\frac{a'}{a}\right| 2\pi = \pi\varepsilon \left|1 - \frac{a^2}{4}\right|.$$

Small net change alone is not sufficient

At $a = 2$ the averaged change is zero because gain and loss cancel. The within-cycle changes can still be large if feedback is strong.

We also require $\varepsilon \ll 1$ and, for large releases, $\varepsilon a^2 \ll 1$. A small net cycle average does not guarantee a nearly sinusoidal waveform.

Use the validity estimate before trusting a transient

For the standard model, a quick estimate is

$$\left|\frac{\Delta a}{a}\right| \approx \pi\varepsilon \left|1 - a^2/4\right|.$$

A weak example: $\varepsilon = 0.01$, $a = 1$

$$\pi(0.01)|1 - 1/4| = 0.0236 : \quad \text{about } 2.36\% \text{ per cycle.}$$

The earlier large release: $\varepsilon = 0.10$, $a = 3$

$$\pi(0.10)|1 - 9/4| = 0.393 : \quad \text{about } 39\% \text{ per cycle.}$$

The latter is not a small fractional change, so its early transient needs numerical checking. The estimate itself freezes amplitude and is only a warning indicator when it is large. Agreement near the final cycle does not guarantee equal accuracy throughout the transient.

Compare leading predictions with the full equation

For $\varepsilon = 0.10$, the first-order prediction is

$$a_* \approx 2, \qquad \omega \approx 1, \qquad T_\tau \approx 2\pi = 6.28319.$$

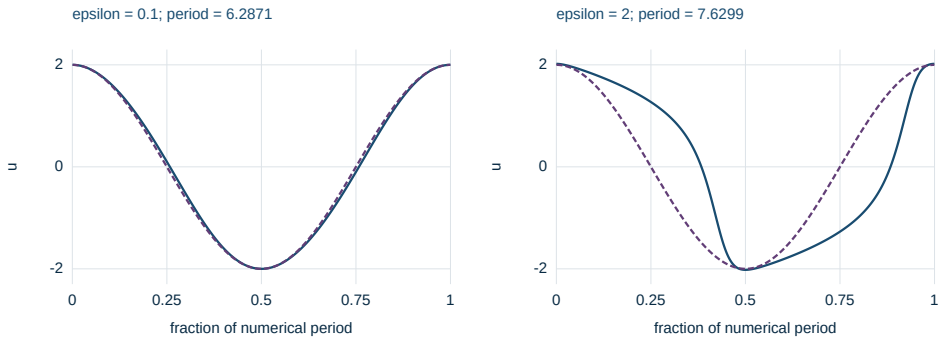
Direct integration of the full Van der Pol equation gives a settled positive peak about 2.00008 and a period about 6.28711. Thus

$$\frac{6.28711 - 6.28319}{6.28319} \times 100\% \approx 0.0624\%.$$

The leading amplitude is very accurate here, while a small period error remains. Zero first-order mean phase drift does not imply zero frequency correction at higher orders.

As ε grows, the waveform itself becomes less sinusoidal. The next plot compares settled cycles at the same fraction of their own periods so their shapes can be compared directly.

Stronger nonlinear damping changes the waveform



Blue: settled numerical cycles; purple dashed: a cosine of amplitude 2, aligned by cycle fraction. At $\varepsilon = 2$ the distortion is substantial; first-order averaging is no longer a quantitatively reliable waveform model.

Prediction check: decay, growth, or sustained motion?

For $u'' + u = 0.10(\mu - u^2)u'$, predict what happens after release from rest:

Case	Initial data	Question
A	$\mu = -0.25$, $u(0) = 0.5$	Can a cycle be sustained?
B	$\mu = 1$, $u(0) = 3$	Does the amplitude grow or decrease?
C	$\mu = 0.25$, $u(0) = 0.2$	What amplitude is approached?
D	$\mu = 1$, $u(0) = u'(0) = 0$	Does motion start without a disturbance?

Use the sign of $a' = 0.05a(\mu - a^2/4)$ and distinguish an unstable equilibrium from a moving trajectory.

Prediction answers: energy loss and a large release

Use $a' = 0.05a(\mu - a^2/4)$ for the stated $\varepsilon = 0.10$.

A: $\mu = -0.25$, initial amplitude 0.5

For every $a > 0$, $-0.25 - a^2/4 < 0$, so amplitude decreases. Initially,

$$a' = 0.05(0.5)(-0.25 - 0.5^2/4) = 0.025(-0.3125) = -0.0078125.$$

There is no nonzero steady amplitude; motion decays to rest.

B: $\mu = 1$, initial amplitude 3

$$a' = 0.05(3)(1 - 3^2/4) = 0.15(-1.25) = -0.1875.$$

The averaged amplitude decreases toward 2. It does not decrease to zero because smaller amplitudes eventually have net energy gain.

These are slow-envelope rates. At the instant of release from rest, exact power and the exact instantaneous radius rate are zero. Averaging describes the ensuing cycle change.

Prediction answers: small release and exact rest

C: $\mu = 0.25$, initial amplitude 0.2

$$a' = 0.05(0.2)(0.25 - 0.2^2/4) = 0.010(0.24) = 0.0024 > 0.$$
$$a_* = 2\sqrt{0.25} = 1.$$

The envelope grows toward 1. Its initial release determines the transient, not the selected amplitude.

D: $\mu = 1$, $u(0) = u'(0) = 0$

The full equation gives $u'' = 0$ at the zero state, and $u(\tau) = 0$ remains an exact solution.

Instability means that a nonzero disturbance grows. It does not generate a disturbance from mathematically exact rest. Physical systems normally have noise or an imperfect release.

What distinguishes a self-excited limit cycle?

Energy balance selects amplitude

Negative damping supplies energy at small motion; nonlinear damping removes it at larger motion. The selected cycle has zero net energy change per period.

The slow flow predicts stability

A stable nonzero amplitude corresponds to attraction toward a periodic orbit. Phase remains free because the system is autonomous.

Hopf describes a local onset

With independent gain control, the equilibrium loses stability and a stable cycle emerges with amplitude proportional to $\sqrt{\mu}$.

Next week: periodic external forcing, primary resonance, and frequency-response curves. Self-excited oscillation and forced resonance have different energy-input mechanisms.

Sources for the methods and examples

[1] A. H. Nayfeh and D. T. Mook

Nonlinear Oscillations. Wiley, 1979.

Secs. 3.1.6 and 3.3.1–3.3.4: negative damping, multiple scales, averaging, linear damping and self-sustained oscillations. Eq. (3.1.36): Van der Pol model. Sec. 1.3: limit-cycle interpretation.

[2] Y. A. Kuznetsov

"Andronov–Hopf bifurcation." *Scholarpedia*, 1(10):1858, 2006. DOI: 10.4249/scholarpedia.1858.

Local eigenvalue crossing, amplitude normal form, and supercritical/subcritical classification.